

RANDOM QUOTIENT-LEAKAGE EROSION: EVERY-TARGET FIBRES AND A COMPLEMENTARY JORDAN LADDER

ANONYMOUS

ABSTRACT. Let $V = \mathbb{F}_q^n$. At every epoch, resample a uniform endomorphism T of V and replace the current subspace U by $U \cap T^{-1}(U)$. Factoring $T|_U$ through V/U identifies the update with the kernel of a uniform quotient map. This gives the exact fibre of every labelled target, an $(n+1)$ -state dimension quotient, and every-time labelled transition probabilities. The full transition operator has algebraic eigenvalues $q^{-a(n-a)}$ with Gaussian multiplicities. In the dimension quotient, each equality between complementary dimensions a and $n-a$ produces a forced 2-by-2 Jordan block, while the two eigenvalues at one remain semisimple because zero and V are separately fixed. Exact absorption formulas complete the finite atlas. Finite-field rank laws, Gaussian incidence, and generic triangular-chain algebra are treated as background; the external status is `HOLD_EXTERNAL`.

1. THE QUOTIENT-KERNEL CHAIN

Fix a prime power q and an n -dimensional vector space V over $\mathbb{F}_q$. Let $T_1, T_2, \dots$ be independent uniform elements of $\text{End}(V)$ and set

$$(1) \quad U_r = U_{r-1} \cap T_r^{-1}(U_{r-1}).$$

This is a Markov chain on the full labelled subspace lattice. The ordinary rank and nullity distributions of uniform finite-field matrices are well developed; see, for example, [1, 2]. Those rank laws, the Gaussian subspace census, and generic finite-chain facts are assigned no contribution credit here. Our narrow object is the state-dependent quotient in (1) and the resulting combination of target fibres, labelled powers, and spectral resonance.

For $0 \leq b \leq a \leq n$, put $d = a - b$ and define

$$(2) \quad C_{n,q}(a, b) = \begin{cases} \prod_{i=0}^{d-1} (q^{n-a} - q^i), & d \leq n - a, \\ 0, & d > n - a. \end{cases}$$

The empty product is one. Write $\begin{bmatrix} a \\ b \end{bmatrix}_q$ for a Gaussian coefficient and let

$$(3) \quad Q_{ab} = \begin{bmatrix} a \\ b \end{bmatrix}_q \frac{C_{n,q}(a, b)}{q^{a(n-a)}} \quad (0 \leq b \leq a \leq n),$$

with all other entries zero. Let P be the full transition matrix indexed by labelled subspaces.

Theorem 1 (labelled fibre and spectral atlas). *The chain (1) has the following exact description.*

(i) *Fix $B \leq U \leq V$ with $\dim U = a$ and $\dim B = b$. Then*

$$(4) \quad \#\{T \in \text{End}(V) : U \cap T^{-1}(U) = B\} = q^{n^2 - a(n-a)} C_{n,q}(a, b).$$

A target not contained in U has empty fibre. Consequently

$$(5) \quad P(U, B) = \frac{C_{n,q}(a, b)}{q^{a(n-a)}}.$$

(ii) *The matrix Q in (3) is the dimension transition matrix. For every $t \geq 0$ and every fixed $B \leq U$,*

$$(6) \quad P^t(U, B) = \frac{(Q^t)_{ab}}{\begin{bmatrix} a \\ b \end{bmatrix}_q}.$$

The probability is zero when $B \not\leq U$.

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(iii) The complete algebraic eigenvalue multiset of P is

$$(7) \quad \left\{ q^{-a(n-a)} \text{ with multiplicity } \begin{bmatrix} n \\ a \end{bmatrix}_q : 0 \leq a \leq n \right\}.$$

The full Jordan form is not asserted. The Jordan form of the dimension quotient Q , however, is completely determined: the eigenvalue one has two J_1 blocks; for every $1 \leq b < n/2$, the common eigenvalue

$$(8) \quad q^{-b(n-b)} = q^{-(n-b)b}$$

has one J_2 block; and when n is even, the middle eigenvalue $q^{-n^2/4}$ has one J_1 block.

(iv) The subspaces 0 and V are fixed. Every proper nonzero starting subspace is absorbed at 0 almost surely. If $0 < a < n$, then

$$(9) \quad \mathbb{P}_a(\tau_0 \leq t) = (Q^t)_{a0},$$

$$(10) \quad E_0 = 0, \quad E_a = \frac{1 + \sum_{b < a} Q_{ab} E_b}{1 - Q_{aa}} = \mathbb{E}_a \tau_0.$$

For $n = 2$ the sole transient row is

$$(11) \quad Q_{10} = \frac{q-1}{q}, \quad Q_{11} = \frac{1}{q}, \quad E_1 = \frac{q}{q-1},$$

while 0 and V remain fixed.

2. EVERY-TARGET FIBRES AND LABELLED POWERS

Let $\pi_U : V \rightarrow V/U$ be the quotient projection. Only the map

$$(12) \quad L_T = \pi_U \circ T|_U : U \longrightarrow V/U$$

is visible to the update, and

$$(13) \quad U \cap T^{-1}(U) = \ker L_T.$$

Proof of Theorem 1(i). The linear map

$$\text{End}(V) \longrightarrow \text{Hom}(U, V/U), \quad T \longmapsto L_T,$$

is surjective. Its codomain has dimension $a(n-a)$, so every quotient map has $q^{n^2-a(n-a)}$ lifts to an ambient endomorphism. Requiring $\ker L_T = B$ is equivalent to requiring the induced map $U/B \rightarrow V/U$ to be injective. There are exactly

$$\prod_{i=0}^{a-b-1} (q^{n-a} - q^i) = C_{n,q}(a, b)$$

such maps, with none when $a-b > n-a$. Multiplying by the common number of lifts proves (4); dividing by q^{n^2} proves (5). $\square$

Summing (5) over the $\begin{bmatrix} a \\ b \end{bmatrix}_q$ subspaces $B \leq U$ of dimension b proves both the row-sum identity for (3) and the dimension quotient.

Proof of Theorem 1(ii). Every trajectory is nested. Moreover, the stabilizer of U in $\text{GL}(V)$ acts transitively on its b -subspaces and preserves the distribution of the freshly sampled endomorphisms. Conditional on starting at U and ending in dimension b , all $\begin{bmatrix} a \\ b \end{bmatrix}_q$ possible labelled endpoints therefore have equal probability. Their total mass is $(Q^t)_{ab}$, giving (6), including $t = 0$. $\square$

Thus the small quotient loses no labelled endpoint information: once Q^t is known, one division recovers every reachable target.

3. COMPLEMENTARY-DIMENSION RESONANCE

Order the labelled subspaces by nondecreasing dimension. The transition matrix is lower triangular because (1) only decreases the state. Remaining at an a -space is equivalent to $L_T = 0$, hence

$$(14) \quad P(U, U) = q^{-a(n-a)}.$$

There are $\binom{n}{a}_q$ such subspaces, proving the algebraic multiset (7).

Lemma 2. *For $1 \leq k \leq n-1$, the adjacent entry $Q_{k,k-1}$ is positive.*

Proof. Here the dimension loss is one and the codomain has dimension $n-k \geq 1$. Thus $C_{n,q}(k, k-1) = q^{n-k} - 1 > 0$ in (3). $\square$

Proof of the quotient Jordan assertion. The exponent $x(n-x)$ is strictly concave and symmetric around $n/2$. Consequently the only repetitions among the diagonal entries of Q are the complementary pairs $b, n-b$, together with the endpoints $0, n$.

Fix $1 \leq b < n/2$, put $a = n-b$, and let $\lambda = q^{-b(n-b)}$. Consider the right-eigenvector recursion $(Q - \lambda I)x = 0$ in increasing row order. Rows below b force $x_0 = \dots = x_{b-1} = 0$, while row b leaves x_b free. Scale a putative eigenvector born there so that $x_b = 1$. For $b < k < a$, strict concavity gives $Q_{kk} < \lambda$, and the recursion reads

$$x_k = \frac{\sum_{j < k} Q_{kj} x_j}{\lambda - Q_{kk}} > 0.$$

The inequality follows inductively from Lemma 2. At row a , however, the diagonal again equals λ , while the compatibility sum is strictly positive because it contains $Q_{a,a-1} x_{a-1} > 0$. Hence no eigenvector can be born at b .

The eigenvector born at a remains, so this algebraically double eigenvalue has geometric multiplicity one and therefore one J_2 block. At eigenvalue one, both 0 and V are isolated fixed rows, giving two independent eigenvectors and two J_1 blocks. If n is even, the midpoint occurs only once and gives one J_1 . These blocks exhaust the $n+1$ quotient dimensions. $\square$

Functions depending only on subspace dimension form a P -invariant subspace represented by Q . Therefore every displayed J_2 is also a genuine obstruction to diagonalizability of the full operator. We do not infer the remaining full-operator Jordan multiplicities from this quotient.

4. ABSORPTION, BOUNDARIES, AND CLAIM BOUNDARY

Proof of Theorem 1(iv). The quotient in (12) is zero when $U = 0$ or $U = V$, so those states are fixed. If $0 < \dim U < n$, there is positive probability that L_T is nonzero, and hence positive probability of strict dimension loss. The finite nested chain consequently reaches zero almost surely. Formula (9) is (6) with the unique zero-dimensional target. First-step conditioning gives (10). Substitution in (3) gives (11). The cases $n = 0$ and $n = 1$ consist only of fixed boundary states. $\square$

The residual claim is intentionally narrower than random-matrix rank theory. Balakin's rank distribution and the finite-field nullity laws studied by Fulman and Goldstein [1, 2] own the matrix counts used in (2); Gaussian incidence owns the number of labelled subspaces; and triangular spectra, Jordan recursions, and hitting-time identities are standard. Internally, earlier subspace-lattice and random-intersection notes own their carriers and generic proof devices. What is retained here is only the literal quotient-kernel process (1), its ambient every-target lift count (4), the all-time labelled recovery (6), and the forced ladder indexed by complementary dimensions.

The standalone verifier independently enumerates every binary subspace and every ambient endomorphism through $n = 3$, checks labelled powers through four epochs, and tests exact rational Jordan nullities for $q \in \{2, 3, 4, 5\}$ through $n = 9$. These computations can falsify arithmetic or boundary errors but do not prove the uniform statements. The source search was bounded; a non-hit is not evidence of novelty, priority, or freedom to operate. The manuscript therefore remains anonymous and `HOLD_EXTERNAL`.

REFERENCES

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